

Unit 8 – Inference for Categorical Data
HW 4 – Chi-Square Test for Association/Independence

Name: _____

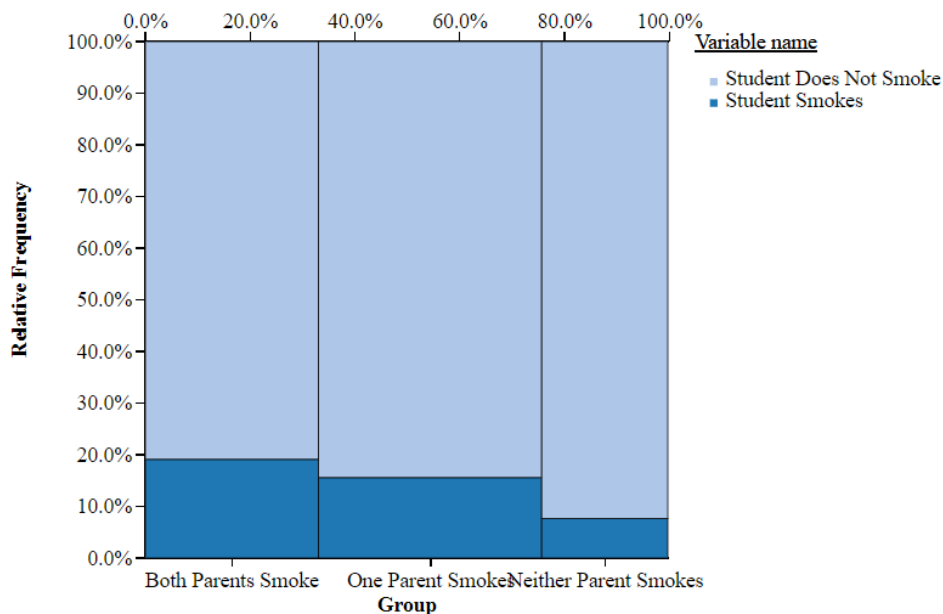
Are the smoking habits of college students related to their parents' smoking habits? Below are data from a survey of students gathered across five different colleges.

	Both Parents Smoke	One Parent Smokes	Neither Parent Smokes
Student Smokes	300	316	88
Student Does Not Smoke	1280	1723	1068

a) How can the data be gathered so that we would perform a chi-square test for homogeneity?

b) How can the data be gathered so that we would perform a chi-square test for association/independence?

c) (Units 2 Throwback) Below is a mosaic plot of the data. In a mosaic plot, the width of each vertical bar represents the relative size of each parent smoking habit category. Are the categories (Both Parents Smoke, One Parent Smokes, Neither Parent Smokes) equally represented in the survey? Explain.



d) What would be the null and alternative hypotheses from a chi-square test for association/independence?

e) Using your calculator, find the expected counts, chi-square statistic, degrees of freedom, and the p-value.

Expected Counts	Both Parents Smoke	One Parent Smokes	Neither Parent Smokes
Student Smokes			
Student Does Not Smoke			

$\chi^2 =$ _____

df = _____

P-value = _____

f) Based on your p-value, what can you conclude?

g) Which cell in the table contributed the most to the chi-square statistic? How does this information expand on your solution in part (f)?